College of Engineering and Technology

Project Work – Student Hand Book

Name of student	Register Number	Department	Mobile Number	Email ID
Yash Chauhan	RA2311026010170	CINTEL	8208952985	Yc0810@srmist.edu.in
Shreya Srivastava	RA2311026010186	CINTEL	8090602858	Ss1371@srmist.edu.in

Title of the Project:	**Digital Inclusion and Circular Economy Behavior in Indian States: A Digital Circularity Index and Explainable XGBoost Analysis**
Project Site / Location	SRM Institute of Science and Technology, Kattankulathur Campus, Tamil Nadu, India
Name and address of the company / organisation (Applicable for projects with industry or industry support)	SRM IST, Kattankulathur, Chengalpattu District – 603203 (in-house academic project)

Supervision Team

	Supervisor	Co-Supervisor	External Supervisor (If applicable)
Name	**Dr. Suresh Kumar**		
Designation	Associate Professor		
Department	Computational Intelligence		
Campus	Kattankulathur		
Telephone			
E-mail	sureshk5@srmist.edu.in		

Degree / Programme	B.Tech	**Specialisation**	Computer Science & Engineering
Academic Year	2025-2026 (Even)	**Semester**	6
Course Code	21CSP302L	**Course Title**	Minor Project

Mission Statement
To deliver a reproducible, evidence-driven study that quantifies the relationship between digital inclusion and circular-economy behavior across 16 Indian states by constructing a Digital Circularity Index (DCI) and validating it through classical statistics, leak-free machine learning, and SHAP-based explainability — so that policy decisions on Digital India and Swachh Bharat can be informed by a single, integrated analytical lens.
Problem / Product Description
India is simultaneously expanding digital infrastructure (tele-density above 85%, UPI exceeding 14 billion monthly transactions) and pursuing circular-economy goals across 11 priority sectors under NITI Aayog's strategy. Yet research on these two agendas has stayed largely separate: Digital India studies focus on growth and financial inclusion, while CE studies focus on waste-management technology and Extended Producer Responsibility.
There is no state-level composite that links digital access with circular-economy outcomes, and no defensible small-n analytical pipeline that pairs OLS with leak-free machine-learning validation. As a result, policy makers cannot say whether connectivity investment also functions as a sustainability investment — a question with direct implications for cross-ministry budget allocation.
This project addresses that gap by integrating seven verified government data sources (TRAI, RBI, NITI Aayog, NPCI, CPCB, Census of India, and Dataful) into a 16-state dataset, constructing three composite indices (Digital Access Score, CE Behavior Score, Digital Circularity Index), and testing three hypotheses (H1 association, H2 UPI mediation, H3 tele-density threshold) using OLS, bootstrap, HC3, permutation, mediation, piecewise regression, seven-model LOO-CV, leak-free nested-CV XGBoost, SHAP explainability and K-Means clustering.
Deliverables: a unified state-level dataset with full provenance, 22 publication-ready figures, an IEEE-format conference paper, a reproducible Python code base, and a defense-ready minor project report.
Assumptions and Constraints
A1. Inner-join across the seven sources yields exactly 16 states with complete observations; no imputation is performed.
A2. Min–max normalisation to [0, 100] and the documented weight schemes (DAS 40/30/30, CEBS 35/40/25) are appropriate for a transparent composite-index construction.
A3. Cross-sectional state-level data supports correlational inference (H1) but not causal claims; mediation (H2) and threshold (H3) results are interpreted with that limit in mind.
C1. n = 16 is small — every model is validated under LOO-CV or nested LOO-CV, and reviewer-defense statistics (HC3, bootstrap 10k, permutation 10k) are mandatory.
C2. Census 2011 is the most recent available source for literacy and urbanisation; downstream interpretation accounts for this lag.
C3. Reproducibility requires fixed random seeds (seed = 42) and saved intermediate CSVs at every pipeline boundary.

Stakeholders
Primary — Project supervisor (Dr. Suresh Kumar) and the Department of Computational Intelligence, SRMIST.
Academic — Project Review Committee, IEEE conference reviewers, and peer researchers in ICT4D and circular-economy analytics.
Policy — NITI Aayog (SDG India Index team), Ministry of Electronics & IT (Digital India), Ministry of Environment (Swachh Bharat / EPR), and state planning departments.
Data Providers — TRAI, RBI, NPCI, CPCB, Census of India, NITI Aayog and Dataful — whose published datasets feed the unified state-level table.
End Users — Policy analysts and researchers using the DCI as a diagnostic tool for cross-ministry planning.

Division of work and contributors of SPRINT 1 [Include Daily Scrum of Sprint 1]

Sprint 1 — Data Engineering, EDA & Statistical Validation

Duration: 01 February 2026 – 21 February 2026 (3 weeks)

Sprint Goal: Build a verified 16-state dataset, compute the DAS / CEBS / DCI composite indices, generate publication-grade EDA, and produce a defense-ready statistical test of H1, H2 and H3.

Yash Chauhan (RA2311026010170) — Contributions

- Source ingestion from TRAI, RBI, NITI Aayog and NPCI (01_data_engineering.py).
- Inner-join merge on the State key; min–max normalisation; construction of DAS, CEBS and DCI columns.
- OLS regression of CEBS on DAS with full diagnostics — Breusch-Pagan, Durbin-Watson, Cook's distance.
- Robustness battery: HC3 robust standard errors, bootstrap CIs (10,000 resamples) and permutation test (10,000 resamples).

Shreya Srivastava (RA2311026010186) — Contributions

- Source ingestion from CPCB, Census of India and Dataful; data-quality QA and provenance logging.
- EDA figures fig1–fig8: correlation heatmap, DAS-vs-CEBS scatter, DCI ranking, distributions, boxplots, UPI trends, regression panels and UPI-vs-SDG.
- Mediation analysis (Baron & Kenny, Sobel) and piecewise threshold regression for H2 / H3.
- Weight-sensitivity analysis across seven alternative DAS schemes; extended-sample (n = 21) validation.

Daily Scrum Highlights — Sprint 1

- 03 Feb — Inner-join produces 16 clean states with 0 missing values; DAS, CEBS and DCI columns committed.
- 09 Feb — Primary OLS confirmed — $r = 0.781$, $R^2 = 0.610$, $p < 0.001$.
- 14 Feb — HC3 $p = 1.1\times10^{-5}$ and bootstrap 95% CI [0.648, 0.979] both exclude 0.
- 18 Feb — Mediation Sobel $z = -0.241$ (no UPI mediation); piecewise breakpoint located at 118.77% tele-density.

- 21 Feb — Sprint 1 retrospective committed; data and statistical artifacts handed off to Sprint 2.

Signature of the Supervisor

Division of work and contributors of SPRINT 2 [Include Daily Scrum of SPRINT 2]

Sprint 2 — Machine Learning, Explainability & Clustering

Duration: 22 February 2026 – 08 March 2026 (~2.5 weeks)

Sprint Goal: Confirm the OLS signal using out-of-sample machine learning, quantify feature importance through SHAP, and uncover state typologies with K-Means clustering.

Yash Chauhan (RA2311026010170) — Contributions

- Seven-model LOO-CV harness — Linear, Ridge, Lasso, ElasticNet, SVR, Random Forest and Gradient Boosting — on digital-only and full feature sets.
- Leak-free nested LOO-CV wrapper for XGBoost — outer LOO with inner 4-fold GridSearchCV.
- Repeated K-Fold cross-validation stability check across 100 × 4 = 400 splits.

Shreya Srivastava (RA2311026010186) — Contributions

- SHAP global and per-state explanations — beeswarm, bar and waterfall plots for all 16 states (focus on Haryana and Bihar).
- K-Means clustering with silhouette-selected k = 2; PCA(2) biplot for geometric interpretation.
- Cluster profile table separating Digital Leaders from Emerging States.

Daily Scrum Highlights — Sprint 2

- 24 Feb — Lasso full-model LOO-CV $R^2 = 0.568$; ElasticNet digital-only $R^2 = 0.364$.
- 28 Feb — Nested-CV XGBoost reaches $R^2 = 0.511$, $p = 0.002$ (leak-free).
- 02 Mar — SHAP global ranks tele-density (5.01) and Per-Capita NSDP (4.88) as the top two drivers.
- 05 Mar — K-Means k = 2 yields a 2.5× DCI gap — Digital Leaders 46.7 vs. Emerging States 18.3.
- 08 Mar — Sprint 2 retrospective; ML and clustering outputs handed to the writing track.

Signature of the Supervisor

Division of work and contributors of SPRINT 3 [Include Daily Scrum of Sprint 3]

Sprint 3 — Documentation, Paper Submission & Review Defense

Duration: 09 March 2026 – 25 April 2026 (~6 weeks)

Sprint Goal: Compile the IEEE-format conference paper, finalise the minor project report, and prepare reviewer-defense materials.

Yash Chauhan (RA2311026010170) — Contributions

- IEEE conference paper draft — Abstract, Introduction, Methods, Results, Discussion, Conclusion — with a full IEEE-style reference list.
- Final minor project report with seven chapters, table of contents, list of figures, list of tables and abbreviations index.
- Reproducibility audit — fixed seeds, saved CSV outputs at every pipeline boundary, README with run instructions.

Shreya Srivastava (RA2311026010186) — Contributions

- 22 publication-ready figures at 300 DPI; figure captions cross-checked against the corresponding results files.
- Reviewer-defense Q&A — small-n justification, multicollinearity (VIF), HC3 vs. OLS, nested-CV vs. flat CV.
- Policy-implications section drawing on SHAP and clustering — Digital Leaders vs. Emerging States playbooks.

Daily Scrum Highlights — Sprint 3

- 14 Mar — Paper Sections 1–3 drafted; figures 1–12 finalised.
- 22 Mar — Paper Sections 4–6 drafted; nested-CV table and SHAP plots embedded.
- 02 Apr — Internal review pass on the paper; minor report TOC, List of Figures and List of Tables generated.
- 17 Apr — Final report ready — page borders, page numbers, black-and-white tables and the architecture diagram added.
- 25 Apr — Handbook compiled, sprint retrospectives archived, submission packet ready.

Signature of the Supervisor

Worksheet / Data collection / Observation etc

A. Verified Primary Data Sources

- TRAI (2024) — Tele-density (%) per state, Quarterly Indicator Report.
- RBI (2024–25) — Per-Capita NSDP (INR), Handbook of Statistics, Table 9.
- NITI Aayog (2023–24) — SDG India Index Score per state.
- NPCI / Dataful (2024–25) — UPI per-capita transactions, monthly product statistics.
- CPCB (2021–22) — E-waste processed (tonnes), Annual e-waste report.
- Census of India (2011) — Literacy rate (%) and Urban population (%), Primary Census Abstract.

B. Primary Outputs of Data Collection

- /data/trai_state_data_final_v2.csv — unified 16-state × 15-variable table with explicit provenance columns.
- /results/ — OLS, robustness, mediation, piecewise, LOO-CV, nested-CV, SHAP and clustering result files.
- /figures/ fig1–fig22 — 22 publication-ready figures rendered at 300 DPI.

C. Descriptive Statistics (n = 16)

- DAS — Mean 64.95, Std. Dev. 19.69, Range 30.27 to 94.40, Median 67.02.
- CEBS — Mean 46.98, Std. Dev. 13.90, Range 27.23 to 87.30, Median 47.78.
- DCI — Mean 32.52, Std. Dev. 17.40, Range 8.24 (Bihar) to 72.97 (Haryana).

D. Headline Observations

- Pearson r(DAS, CEBS) = 0.781, $p < 0.001$ — strong positive association.
- OLS $R^2 = 0.610$; HC3 robust $p = 1.1 \times 10^{-5}$; bootstrap 95% CI for r = [0.648, 0.979].
- Tele-density tops SHAP global importance (mean $|\varphi| = 5.01$); UPI ranks last (0.91), consistent with the non-significant Sobel mediation ($z = -0.241$).
- K-Means k = 2 separates Digital Leaders (DCI 46.7) from Emerging States (DCI 18.3) — a 2.5× gap.

Research Article with Journal Publication Details / Patent disclosure form with patent status

(includes certificates and proofs)

Title:

Digital Inclusion and Circular Economy Behavior in Indian States: A Digital Circularity Index and Explainable XGBoost Analysis.

Authors:

Yash Chauhan (RA2311026010170), Shreya Srivastava (RA2311026010186); Supervisor — Dr. Suresh Kumar.

Affiliation:

Department of Computational Intelligence, SRM Institute of Science and Technology, Kattankulathur, Tamil Nadu, India.

Target Venue:

IEEE conference (paper formatted to the IEEE conference template).

Status:

Manuscript drafted; figures 1–22 embedded; reference list complete; submission package prepared (paper24.pdf, paper24.tex).

Patent Disclosure:

Not applicable.

Supporting Artifacts:

Defense-ready minor project report, reproducible Python code base, and certificates / proofs attached separately.